

WTW158 Calculus

Class Test 1 (Mo)

17 February - 21 February 2025

Time: 15 min

8

Surname:

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Initials:

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Student number:

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Group:

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Use a pen to complete the test. Show all your steps and simplify your answers. You are not allowed to use a calculator. The test is printed on both pages.

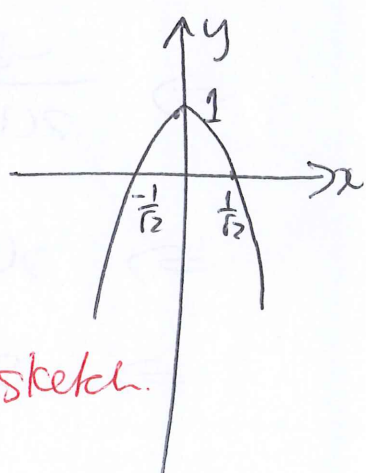
QUESTION 1 [3 marks]

Rewrite the given expression using the definition of the absolute value. Use interval notation where applicable.

$$|1-2x^2| = \begin{cases} 1-2x^2 & \text{if } 1-2x^2 \geq 0 \\ -(1-2x^2) & \text{if } 1-2x^2 < 0 \end{cases} \quad \text{Definition of abs. value.}$$

$$= \begin{cases} 1-2x^2 & \text{if } x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] \\ -1+2x^2 & \text{if } x \in (-\infty, -\frac{1}{\sqrt{2}}) \cup (\frac{1}{\sqrt{2}}, \infty) \end{cases}$$

$1-2x^2=0 \Rightarrow 1=2x^2 \Rightarrow x=\pm\frac{1}{\sqrt{2}}$



From sketch.

QUESTION 2 [2 marks]

Solve the given inequality and write your answer in interval notation.

$$|3x-7| < 8$$

$$-8 < 3x-7 < 8$$

$$-1 < 3x < 15$$

$$-\frac{1}{3} < x < \frac{15}{3}$$

$$\Rightarrow x \in \left(-\frac{1}{3}, 5\right).$$

Interval notation requested.

NB applying theorem

QUESTION 3 [1 mark]

Fully simplify the given expression and write only your final answer in the space provided.

If $a < 0 < b$, then $\left| \frac{a}{b} \right| = \boxed{\frac{-a}{b}}$ and $|a|^3 = \boxed{-a^3}$

QUESTION 4 [2 marks]

Solve the given inequality. Write your final answer in interval notation.

$$\frac{x}{x+2} < 1$$

$$\frac{x}{x+2} - 1 < 0$$

$$\Rightarrow \frac{x - (x+2)}{x+2} < 0$$

$$\Rightarrow \frac{-2}{x+2} < 0$$

$$\Rightarrow x+2 > 0$$

$$\Rightarrow x > -2$$

$$\Rightarrow \underline{x \in (-2, \infty)}$$

MEMO

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QUESTION 1 [2, 3 marks]

Solve the given inequality and write your answer in interval notation.

(i) $|3x + 5| > 1$

$3x + 5 > 1$ OR

$3x + 5 < -1$

$3x > 1 - 5$

$3x > -4$

$x > -4/3$

$3x < -6$

$x < -6/3 = -2$

KB.
Applying the theorem

$x \in (-\infty, -2) \cup (-4/3, \infty)$

(ii) $\frac{2x-1}{x} \leq 3$

$\frac{2x-1}{x} \leq 3$

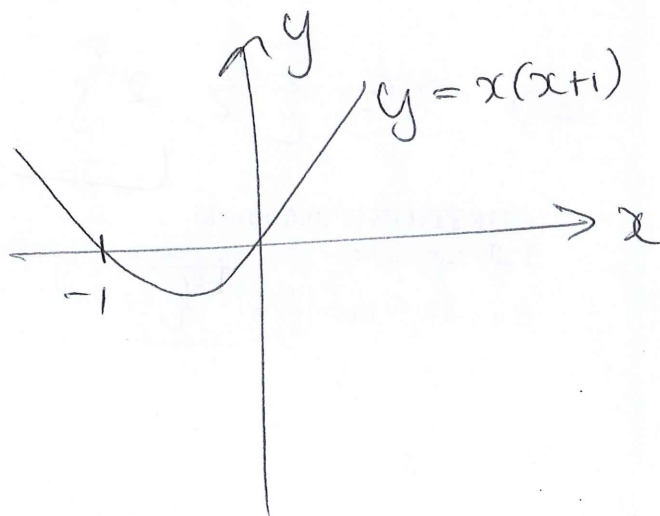
$\Rightarrow \frac{(2x-1)x^2}{x} \leq 3x^2, x \neq 0$

$\Rightarrow (2x-1)x \leq 3x^2, x \neq 0$

$\Rightarrow 2x^2 - x - 3x^2 \leq 0$

$\Rightarrow -x^2 - x \leq 0$

$\Rightarrow x^2 + x \geq 0$



$\Rightarrow$ From sketch:

$x \in (-\infty, -1] \cup (0, \infty)$

QUESTION 3 [1 mark]

Rewrite the given expression without using the absolute value symbol. Use interval notation where applicable.

$$|x - 2| \text{ if } x < 2$$

$$|x - 2| = \begin{cases} x - 2 & \text{if } x - 2 \geq 0 \\ -(x - 2) & \text{if } x - 2 < 0 \end{cases}$$

$$\Rightarrow \text{For } x < 2: \underline{\underline{|x - 2| = -x + 2.}}$$

QUESTION 4 [1.5 marks]

Solve the equation and write your answer as a set.

$$|4x - 5| = 3$$

$$4x - 5 = \pm 3$$

$$\Rightarrow 4x - 5 = 3 \quad \text{or}$$

$$4x - 5 = -3$$

$$\Rightarrow 4x = 8 \quad \text{or}$$

$$4x = -3 + 5 = 2$$

$$\Rightarrow x = 2 \quad \text{or}$$

$$x = \frac{1}{2}$$

} NB: applying theorem

$$\Rightarrow x \in \left\{ \frac{1}{2}, 2 \right\}$$

Set notation requested.

QUESTION 5 [0.5 mark]

Fully simplify the given expression and write only your final answer in the space provided.

If $a < b < 0$, then $\left| \frac{a}{b} \right| = \boxed{\frac{+a}{b}}$



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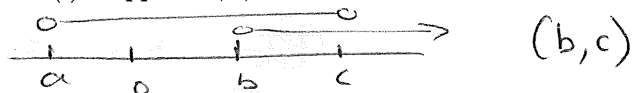
Use a pen to complete the test. Fully simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1 mark]

Indicate whether the statement is true (T) or false (F).

- (i) Suppose a, b, c are real numbers such that $a < 0 < b < c$. Then $(b, \infty) \cap (a, c) = [b, c]$

False



- (ii) The equation $|x^2 - 4| = 3$ has two possible solutions.

False

$$x^2 - 4 = \pm 3$$

$$x^2 = 7 \text{ or } x^2 = 1$$

QUESTION 2 [2 marks]

Fully simplify the given expression and write your final answer in the space provided.

- (i) If $a < 0 < b$, then $\sqrt{(ab)^2} = \boxed{-ab}$

$$\sqrt{(ab)^2} = |ab|$$

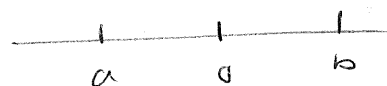
$$= -ab$$

- (ii) If $x < 0$, then $x - |x| = \boxed{2x}$

$$|x| = -x \text{ if } x < 0$$

$$x - |x| = x - (-x)$$

$$= 2x$$



QUESTION 3 [2 marks]

Show how you use the definition of the absolute value function to rewrite the given expression as a piecewise defined function. Use interval notation where applicable.

$$|x^2 - 4| = \begin{cases} x^2 - 4 & \text{if } x^2 - 4 \geq 0 \\ -(x^2 - 4) & \text{if } x^2 - 4 < 0 \end{cases}$$

Definition of abs. value

$$= \begin{cases} x^2 - 4 & \text{if } x \in (-\infty, -2] \cup [2, \infty) \\ -x^2 + 4 & \text{if } x \in (-2, 2) \end{cases}$$

QUESTION 4 [3 marks]

Solve the given inequality, and write your answer in interval notation.

$$\frac{3x+5}{x+1} \leq 2$$

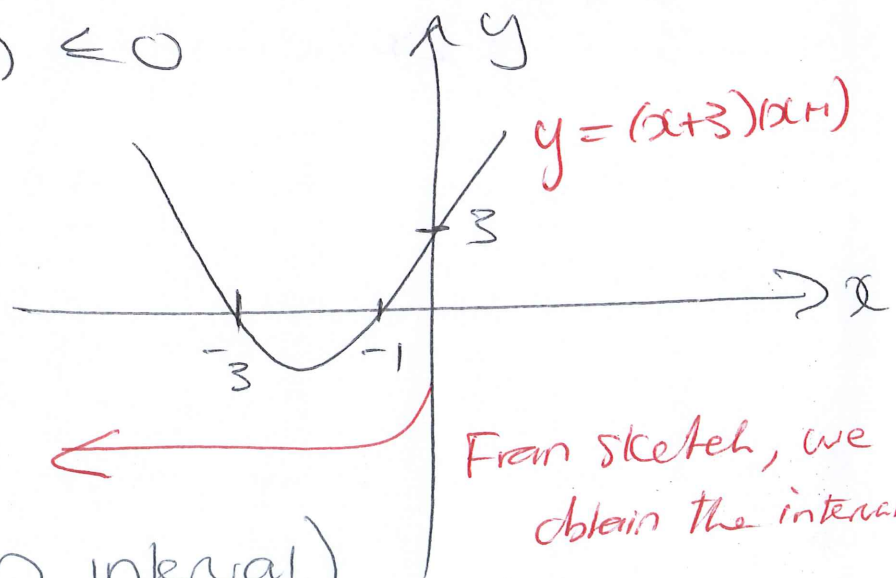
$$\frac{3x+5}{x+1} (x+1)^2 \leq 2(x+1)^2$$

$$\Rightarrow (3x+5)(x+1) \leq 2(x+1)^2, \quad x+1 \neq 0$$

$$\Rightarrow (3x+5 - 2(x+1))(x+1) \leq 0, \quad x \neq -1$$

$$\Rightarrow (3x+5 - 2x - 2)(x+1) \leq 0$$

$$\Rightarrow (x+3)(x+1) \leq 0$$



$$\Rightarrow x \in [-3, -1)$$

(↑ open interval)

From sketch, we obtain the interval

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Use a pen to complete the test. Fully simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1, 0.5 marks]

Indicate whether the statement is true (T) or false (F).

- (i) Suppose S and T are two non-empty sets. The set $S \cap T$ consists of the elements that are in either S or T .

False.

- (ii) Consider $a \in \mathbb{R}$, if $x > a$, then the distance between x and a is $x - a$.

TRUE

$$|x-a| = \begin{cases} x-a & \text{if } x \geq a \\ -(x-a) & \text{if } x < a \end{cases}$$

QUESTION 2 [1.5 marks]

Solve the equation and write your answer as a set.

$$|2x - 3| = 1$$

$$2x - 3 = \pm 1$$

$$\Rightarrow 2x - 3 = 1 \quad \text{or}$$

$$2x = 4 \quad \text{or}$$

$$x = 2 \quad \text{or}$$

$$2x - 3 = -1$$

$$2x = 2$$

$$x = 1$$

NB

From theorem

$$x \in \{1, 2\}.$$

QUESTION 4 [3 marks]

Solve the given inequality, and write your answer in interval notation.

$$\frac{2x+1}{x+1} \geq 3$$

$$\frac{(2x+1)}{(x+1)} (x+1)^2 \geq 3(x+1)^2$$

$$(2x+1)(x+1) \geq 3(x+1)^2, \quad x+1 \neq 0$$

$$\Rightarrow (2x+1-3(x+1))(x+1) \geq 0, \quad x+1 \neq 0$$

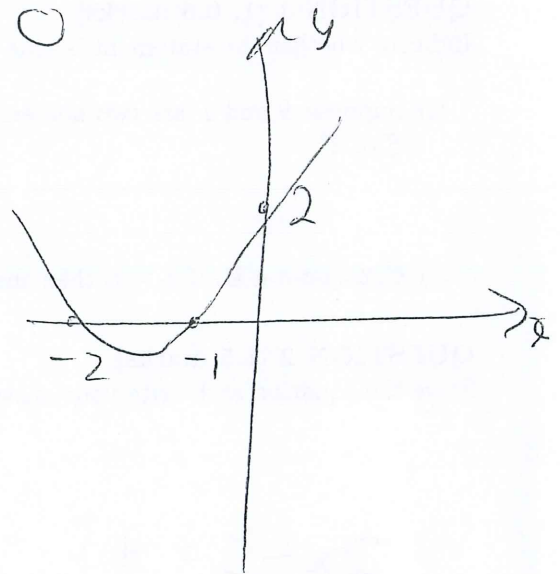
$$\Rightarrow (2x+1-3x-3)(x+1) \geq 0$$

$$\Rightarrow (-x-2)(x+1) \geq 0$$

$$\Rightarrow (x+2)(x+1) \leq 0$$

$$x \in [-2, -1)$$

since $x \neq -1$.



QUESTION 5 [2 marks]

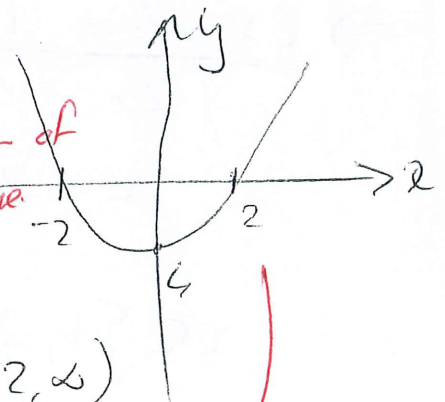
Show how you use the definition of the absolute value function to rewrite the given expression as a piecewise defined function. Use interval notation where applicable.

$$|x^2 - 4|$$

$$|x^2 - 4| = \begin{cases} x^2 - 4 & \text{if } x^2 - 4 \geq 0 \\ -(x^2 - 4) & \text{if } x^2 - 4 < 0 \end{cases}$$

$$= \begin{cases} x^2 - 4 & \text{if } x \in (-\infty, -2] \cup [2, \infty) \\ -x^2 + 4 & \text{if } x \in (-2, 2) \end{cases}$$

Definition of
abs. value



Obtain this
answer from
sketch.